

MAT1002 Final Examination (May 16, 2022)

Sample Solutions

1. True/False

- (i) **T.** Since $D = [4, 8] \times [3, 4] \times [-1, 1]$ is closed and bounded, it is compact. The function

$$f(x, y, z) = xy^2z^6 - \sin(e^{yz}) - \ln(x^2)$$

is continuous on D , so it attains both a maximum and a minimum on D .

- (ii) **F.** Differentiability at a point does not imply continuity of the partial derivatives at that point.

- (iii) **F.** A counterexample is $u_n = \frac{(-1)^{n+1}}{n}$, for which $\sum u_n$ converges, but

$$u_{2n-1} - u_{2n} = \frac{1}{2n-1} + \frac{1}{2n},$$

and hence $\sum(u_{2n-1} - u_{2n})$ diverges.

2. Short questions

- (i)

$$\nabla f = (2x + 2y, 2x + z, y), \quad \operatorname{div}(\nabla f) = f_{xx} + f_{yy} + f_{zz} = 2.$$

- (ii) If

$$f(x, y, z) = \frac{1}{\sqrt{4 - x^2 - y^2 - z^2}} = c,$$

then

$$x^2 + y^2 + z^2 = 4 - \frac{1}{c^2}.$$

So the level surfaces are spheres centered at the origin with radius

$$\sqrt{4 - \frac{1}{c^2}}, \quad c \geq \frac{1}{2}.$$

- (iii) Along the curve C ,

$$\left. \frac{d}{dt} H(x(t), y(t), z(t), t) \right|_{t=0} = f_x x'(0) + f_y y'(0) + f_z z'(0) + f_t$$

evaluated at $(x(0), y(0), z(0), 0)$.

- (iv) For $y = \sin x$,

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{3/2}}.$$

At $x = \frac{\pi}{2}$, we have $y' = \cos \frac{\pi}{2} = 0$ and $y'' = -\sin \frac{\pi}{2} = -1$, so

$$\kappa = 1.$$

(v) Parametrize by $\mathbf{r}(x) = \langle x, -x^2 \rangle$. Then

$$y' = -2x, \quad y'' = -2.$$

The principal unit normal for a graph is

$$\mathbf{N} = \frac{(y', -1)}{\sqrt{1 + (y')^2}} \quad \text{when } y'' < 0.$$

At $x = \frac{1}{2}$, $y' = -1$, so

$$\mathbf{N} = \frac{\langle -1, -1 \rangle}{\sqrt{2}}.$$

(vi)

$$\frac{x^2}{\sqrt{2+x}} = \frac{x^2}{\sqrt{2}} \left[1 + \frac{x}{2} \right]^{-1/2}.$$

Using

$$(1+u)^{-1/2} = 1 - \frac{1}{2}u + \frac{3}{8}u^2 + \cdots,$$

with $u = \frac{x}{2}$,

$$\frac{x^2}{\sqrt{2+x}} = \frac{x^2}{\sqrt{2}} - \frac{1}{4\sqrt{2}}x^3 + \frac{3}{32\sqrt{2}}x^4 + \cdots$$

(vii)

$$H = 5x^2 - 3xy + xyz, \quad \nabla H = (10x - 3y + yz, -3x + xz, xy).$$

At $(3, 4, 5)$,

$$\nabla H(3, 4, 5) = (38, 6, 12).$$

Therefore the direction of fastest decrease is

$$-\nabla H(3, 4, 5) = \langle -38, -6, -12 \rangle.$$

(viii) In spherical coordinates,

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

The spheres

$$x^2 + y^2 + z^2 = z, \quad x^2 + y^2 + z^2 = 2z$$

become

$$\rho = \cos \phi, \quad \rho = 2 \cos \phi.$$

Hence

$$\iiint_E z \, dV = \int_0^{2\pi} \int_0^{\pi/2} \int_{\cos \phi}^{2 \cos \phi} (\rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

(ix) The curve C is a great circle of the unit sphere, so its length is 2π . By symmetry on the plane $x + y + z = 0$,

$$\int_C x^2 \, ds = \int_C y^2 \, ds = \int_C z^2 \, ds.$$

Since on the unit sphere,

$$x^2 + y^2 + z^2 = 1,$$

we get

$$3 \int_C x^2 ds = \int_C 1 ds = 2\pi.$$

Thus

$$\int_C x^2 ds = \frac{2\pi}{3}.$$

3. Consider the surface

$$S : xz^2 - yz + \cos(xy) = 1.$$

(i) Let

$$F(x, y, z) = xz^2 - yz + \cos(xy) - 1.$$

Then

$$\nabla F = (z^2 - y \sin(xy), -z - x \sin(xy), 2xz - y).$$

At $P = (0, 0, 1)$,

$$\nabla F(0, 0, 1) = (1, -1, 0).$$

Therefore the tangent plane is

$$(1, -1, 0) \cdot \langle x, y, z - 1 \rangle = 0 \implies x - y = 0.$$

So

$$M : x - y = 0.$$

A normal line is

$$\ell : (x, y, z) = (0, 0, 1) + t(1, -1, 0),$$

i.e.

$$x = t, \quad y = -t, \quad z = 1.$$

(ii)

$$\mathbf{r}(t) = \langle \ln t, t \ln t, t \rangle.$$

At $t = 1$,

$$\mathbf{r}(1) = \langle 0, 0, 1 \rangle = P.$$

Also

$$\mathbf{r}'(t) = \left\langle \frac{1}{t}, \ln t + 1, 1 \right\rangle, \quad \mathbf{r}'(1) = \langle 1, 1, 1 \rangle.$$

The normal vector of M is $\langle 1, -1, 0 \rangle$, and

$$\langle 1, -1, 0 \rangle \cdot \langle 1, 1, 1 \rangle = 1 - 1 + 0 = 0.$$

Hence the tangent vector is parallel to the plane M , and since $P \in M$, the tangent line lies on M .

4. Let

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

We test the limit at the origin along two paths.

Along $y = 0$,

$$f(x, 0) = 0.$$

Along $y = x$ with $x \neq 0$,

$$f(x, x) = \frac{x^4}{x^4 + 0} = 1.$$

Since the two path limits are different, the limit does not exist. Therefore f is **not continuous** at $(0, 0)$.

5. We want the global extrema of

$$f(x, y, z) = x + y + z$$

subject to

$$x^2 + z^2 = 2, \quad x + y = 1.$$

From $x + y = 1$, we get $y = 1 - x$. Hence

$$f(x, y, z) = x + (1 - x) + z = 1 + z.$$

So we only need to maximize/minimize z subject to

$$x^2 + z^2 = 2.$$

Thus

$$-\sqrt{2} \leq z \leq \sqrt{2}.$$

Therefore:

$$\begin{aligned} f_{\max} &= 1 + \sqrt{2} \quad \text{at } (x, y, z) = (0, 1, \sqrt{2}), \\ f_{\min} &= 1 - \sqrt{2} \quad \text{at } (x, y, z) = (0, 1, -\sqrt{2}). \end{aligned}$$

6. Evaluate the integrals.

(i)

$$\int_0^1 \int_{3y}^3 e^{x^2} dx dy.$$

Reverse the order. The region is

$$0 \leq y \leq 1, \quad 3y \leq x \leq 3,$$

equivalently

$$0 \leq x \leq 3, \quad 0 \leq y \leq \frac{x}{3}.$$

Hence

$$\int_0^1 \int_{3y}^3 e^{x^2} dx dy = \int_0^3 \int_0^{x/3} e^{x^2} dy dx = \int_0^3 \frac{x}{3} e^{x^2} dx.$$

Let $u = x^2$, $du = 2x dx$. Then

$$\int_0^3 \frac{x}{3} e^{x^2} dx = \frac{1}{6} \int_0^9 e^u du = \frac{1}{6}(e^9 - 1).$$

(ii)

$$\int_0^1 \int_0^1 v(u + v^2)^4 du dv.$$

Integrate with respect to u :

$$\int_0^1 v \left[\frac{(u + v^2)^5}{5} \right]_{u=0}^{u=1} dv = \frac{1}{5} \int_0^1 v ((1 + v^2)^5 - v^{10}) dv.$$

Let $w = v^2$, so $dw = 2v dv$:

$$= \frac{1}{10} \int_0^1 ((1 + w)^5 - w^5) dw.$$

Therefore

$$= \frac{1}{10} \left[\frac{(1 + w)^6}{6} - \frac{w^6}{6} \right]_0^1 = \frac{1}{10} \left(\frac{64 - 1}{6} - \frac{1}{6} \right) = \frac{31}{30}.$$

7. The region is inside

$$(x - 1)^2 + y^2 = 1$$

and outside

$$x^2 + y^2 = 1.$$

In polar coordinates, the first circle becomes

$$(r \cos \theta - 1)^2 + r^2 \sin^2 \theta = 1 \implies r = 2 \cos \theta.$$

The second circle is $r = 1$.

Thus the required region is

$$1 \leq r \leq 2 \cos \theta,$$

which exists when $2 \cos \theta \geq 1$, i.e.

$$-\frac{\pi}{3} \leq \theta \leq \frac{\pi}{3}.$$

Hence the area is

$$A = \int_{-\pi/3}^{\pi/3} \int_1^{2 \cos \theta} r dr d\theta = \frac{1}{2} \int_{-\pi/3}^{\pi/3} (4 \cos^2 \theta - 1) d\theta.$$

Using symmetry,

$$A = \int_0^{\pi/3} (4 \cos^2 \theta - 1) d\theta = \int_0^{\pi/3} (1 + 2 \cos 2\theta) d\theta = [\theta + \sin 2\theta]_0^{\pi/3}.$$

Therefore

$$A = \frac{\pi}{3} + \frac{\sqrt{3}}{2}.$$

8. Find the volume enclosed by

$$(x^2 + y^2)^2 + z^2 = 1.$$

In cylindrical coordinates, $x^2 + y^2 = r^2$, so the surface becomes

$$r^4 + z^2 = 1.$$

Hence

$$-\sqrt{1-r^4} \leq z \leq \sqrt{1-r^4}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

So

$$V = \int_0^{2\pi} \int_0^1 \int_{-\sqrt{1-r^4}}^{\sqrt{1-r^4}} r \, dz \, dr \, d\theta = 4\pi \int_0^1 r\sqrt{1-r^4} \, dr.$$

Let $u = r^2$, so $du = 2r \, dr$:

$$V = 2\pi \int_0^1 \sqrt{1-u^2} \, du.$$

But

$$\int_0^1 \sqrt{1-u^2} \, du = \frac{\pi}{4},$$

so

$$V = 2\pi \cdot \frac{\pi}{4} = \frac{\pi^2}{2}.$$

9. Suppose

$$\mathbf{F} = (e^{kx} \ln y)\mathbf{i} + \left(\frac{e^{kx}}{y} + \sin z\right)\mathbf{j} + (my \cos z)\mathbf{k}$$

is conservative on $\{(x, y, z) : y > 0\}$.

(i) Write

$$M = e^{kx} \ln y, \quad N = \frac{e^{kx}}{y} + \sin z, \quad P = my \cos z.$$

For $\mathbf{F}$ to be conservative, we need

$$M_y = N_x, \quad M_z = P_x, \quad N_z = P_y.$$

Compute:

$$M_y = \frac{e^{kx}}{y}, \quad N_x = \frac{ke^{kx}}{y}.$$

Hence $k = 1$.

Also

$$N_z = \cos z, \quad P_y = m \cos z,$$

so $m = 1$.

Therefore

$$k = 1, \quad m = 1.$$

(ii) With $k = m = 1$,

$$\mathbf{F} = (e^x \ln y)\mathbf{i} + \left(\frac{e^x}{y} + \sin z\right)\mathbf{j} + (y \cos z)\mathbf{k}.$$

Let ϕ be a potential. Then

$$\phi_x = e^x \ln y.$$

Integrating with respect to x ,

$$\phi = e^x \ln y + g(y, z).$$

Differentiate with respect to y :

$$\phi_y = \frac{e^x}{y} + g_y(y, z) = \frac{e^x}{y} + \sin z.$$

So

$$g_y = \sin z.$$

Hence

$$g(y, z) = y \sin z + h(z).$$

Now

$$\phi_z = y \cos z + h'(z) = y \cos z,$$

so $h'(z) = 0$. Thus one potential is

$$\boxed{\phi(x, y, z) = e^x \ln y + y \sin z.}$$

10. (i) By Green's theorem, for $M = 0$ and $N = x$,

$$\oint_C x \, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \iint_R 1 \, dA = A(R).$$

Therefore

$$A(R) = \oint_C x \, dy.$$

(ii) The curve is given in polar form by

$$r = \sqrt{\sin \theta}, \quad 0 \leq \theta \leq \pi,$$

since $\sin \theta \geq 0$ on $[0, \pi]$. A parametrization is

$$x(\theta) = \sqrt{\sin \theta} \cos \theta, \quad y(\theta) = \sqrt{\sin \theta} \sin \theta, \quad 0 \leq \theta \leq \pi.$$

(iii) Using part (i),

$$A = \oint_C x \, dy.$$

Here

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{\sin \theta}.$$

Then

$$dy = (r' \sin \theta + r \cos \theta) d\theta.$$

So

$$x dy = r \cos \theta (r' \sin \theta + r \cos \theta) d\theta.$$

Since

$$r' = \frac{\cos \theta}{2\sqrt{\sin \theta}}, \quad rr' = \frac{\cos \theta}{2}, \quad r^2 = \sin \theta,$$

we get

$$x dy = \left(\frac{1}{2} \sin \theta \cos^2 \theta + \sin \theta \cos^2 \theta \right) d\theta = \frac{3}{2} \sin \theta \cos^2 \theta d\theta.$$

Therefore

$$A = \int_0^\pi \frac{3}{2} \sin \theta \cos^2 \theta d\theta.$$

Let $u = \cos \theta$, $du = -\sin \theta d\theta$:

$$A = \frac{3}{2} \int_{-1}^1 u^2 du = \frac{3}{2} \cdot \frac{2}{3} = 1.$$

Hence the enclosed area is

$$\boxed{1}.$$

11. Let

$$S : x = \sqrt{y^2 + z^2}, \quad x \leq 2.$$

We need

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} d\sigma$$

with unit normals pointing in the positive x -direction.

By Stokes' theorem,

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} d\sigma = \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

The boundary is the circle at $x = 2$:

$$C : y^2 + z^2 = 4, \quad x = 2.$$

A compatible parametrization is

$$\mathbf{r}(\theta) = \langle 2, 2 \cos \theta, 2 \sin \theta \rangle, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$\mathbf{r}'(\theta) = \langle 0, -2 \sin \theta, 2 \cos \theta \rangle.$$

Given

$$\mathbf{F} = (\sin(x^3 y^2 z))\mathbf{i} + (x^2 y)\mathbf{j} + (x^2 z^2)\mathbf{k},$$

on C we have

$$x = 2, \quad y = 2 \cos \theta, \quad z = 2 \sin \theta,$$

so

$$\mathbf{F}(\mathbf{r}(\theta)) = \langle \sin(8y^2z), 4y, 4z^2 \rangle = \langle \sin(64 \cos^2 \theta \sin \theta), 8 \cos \theta, 16 \sin^2 \theta \rangle.$$

Thus

$$\mathbf{F} \cdot d\mathbf{r} = (8 \cos \theta) (-2 \sin \theta) d\theta + (16 \sin^2 \theta) (2 \cos \theta) d\theta.$$

Hence

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-16 \sin \theta \cos \theta + 32 \sin^2 \theta \cos \theta) d\theta = 0.$$

Therefore

$$\boxed{0}.$$

12. Let E be the solid above the cone

$$z = \sqrt{\frac{x^2 + y^2}{3}}$$

and below the sphere

$$x^2 + y^2 + z^2 = 1.$$

Let $S = \partial E$. We seek

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma.$$

By the Divergence Theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma = \iiint_E \nabla \cdot \mathbf{F} dV.$$

Given

$$\mathbf{F} = (\cos(z^2))\mathbf{i} + (z^3(y+x))\mathbf{j} + (e^x + yx)\mathbf{k},$$

we have

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(\cos(z^2)) + \frac{\partial}{\partial y}(z^3(y+x)) + \frac{\partial}{\partial z}(e^x + yx) = 0 + z^3 + 0 = z^3.$$

Use spherical coordinates:

$$z = \rho \cos \phi, \quad dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$

The cone is

$$z = \frac{r}{\sqrt{3}},$$

so

$$\tan \phi = \frac{r}{z} = \sqrt{3} \implies \phi = \frac{\pi}{3}.$$

Since the region is above the cone and inside the sphere,

$$0 \leq \rho \leq 1, \quad 0 \leq \phi \leq \frac{\pi}{3}, \quad 0 \leq \theta \leq 2\pi.$$

Therefore

$$\iiint_E z^3 dV = \int_0^{2\pi} \int_0^{\pi/3} \int_0^1 (\rho \cos \phi)^3 \rho^2 \sin \phi d\rho d\phi d\theta.$$

So

$$= \int_0^{2\pi} d\theta \int_0^{\pi/3} \cos^3 \phi \sin \phi d\phi \int_0^1 \rho^5 d\rho = 2\pi \left(\frac{15}{64} \right) \left(\frac{1}{6} \right).$$

Hence

$$\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma = \frac{5\pi}{64}.$$

So the outward flux is

$$\boxed{\frac{5\pi}{64}}.$$